



URBAN REDEVELOPMENT AUTHORITY OF PITTSBURGH

LandCare Assurance

Owner Guide

Prepared for the incoming owner, [ura-gis/land-care-assurance](#)

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1 What you own

LandCare Assurance answers one question for URA supervisors: for this service period, which assigned parcels were actually serviced, by whom, and with what evidence. It is a map-first web application published from this repository, reading live ArcGIS layers, backed by a nightly job on the GIS VM.

Before it existed, contractor compliance was judged from a reported completion figure that blended Active and Request Only parcels against a parcel universe with no ownership check. The application replaces that single percentage with a map, a stated denominator, and a photo per parcel.

1.1 Live routes

All under <https://ura-gis.github.io/land-care-assurance/>.

Route	Audience	Purpose
/monitoring/	Supervisors	Parcel map, evidence, field notes, filters
/kpi/	Leadership	Completion trend, budget, contractor exposure
/contractor/	Contractors	Progress view and prefilled service submission
/survey-submission/	Legacy links	Redirect to /contractor/
/design-system/example.html	Developers	Portable BI kit reference page

1.2 Service period convention

A LandCare period runs the 15th of one month through the 14th of the next, stored as the 15th of the start month. 2026-06-15 means the June to July period. Every count in the product is scoped to a period. When a number looks wrong, check the selected period first.

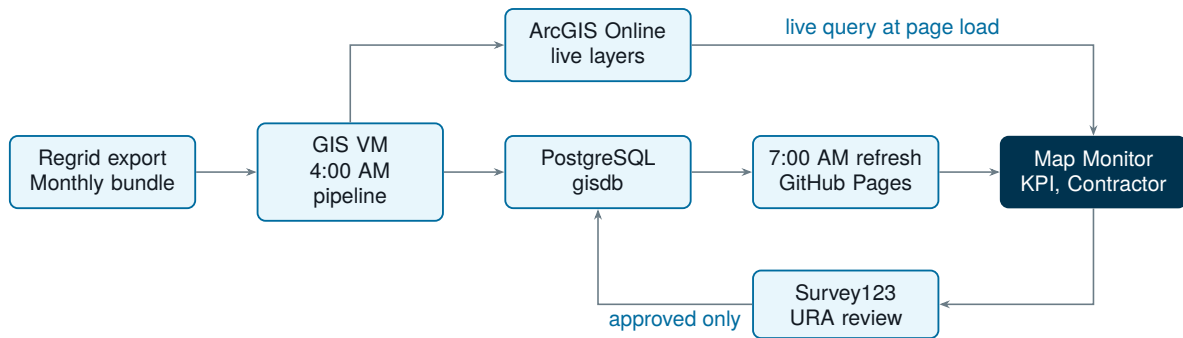
2 System architecture and data flow

Three layers with three different owners.

Layer	Where it runs	Owner
Ingestion	GIS VM Task Scheduler, repo URA-GIS-User/URA-Data-Repository	GIS and data operations
Store	PostgreSQL gisdb and ArcGIS Online (urap.maps.arcgis.com)	GIS and data operations
Publish and application	This repository, GitHub Pages	Web and dashboard owner

2.1 The rule that explains most confusion

Live ArcGIS is authoritative at page load. The JSON and GeoJSON committed under docs/landcare/data/ is a fallback cache plus the finance contract. A number on the map can change during the day with no commit, because the browser queried ArcGIS directly. Do not treat the committed files as truth.



Nightly ingestion publishes to PostgreSQL and ArcGIS Online. The 7:00 AM refresh commits the published data contract and deploys Pages. The browser queries ArcGIS directly at page load, which is why counts move between publishes.

2.2 Daily schedule, Eastern time

Time	Task	Output
4:00 AM	regrid_survey_daily_pipeline.py upstream	Regrid CSV into gis.regrid_survey_submissions, then the ArcGIS survey layer
4:15 AM on the 15th	regrid_survey_monthly_export.py	G-drive CSV archive only, not a dashboard source
7:00 AM	refresh_landcare_dashboard.ps1 in this repo	docs/landcare/data/* committed and pushed when changed

The 7:00 AM job runs three hours after ingestion so the Postgres export reflects the latest state. Because the browser also queries ArcGIS at page load, completion counts can rise between publishes.

2.3 Live ArcGIS sources

All on services1.arcgis.com/0DMNBNaacQNEfn4H.

Layer	Item ID	Used for
gisdb_gis_regrid_surveys	7a2e1d9bacba461296c54a63f104cf51	Evidence, photos, comments, service dates
..._bundle_assignments_current_period	0b4733cb5d204da6ab936c9f6d49e401	Current assignment reference
..._bundle_assignments_history	df7d77eb57f14c68b717c2cf3cdaada4	Monthly assignment denominator
gisdb_gis_epp_parcel_full	live layer	Current URA-owned parcel universe
CouncilDistricts2022	reference	District highlight and filter
LandCare_Survey123_Evidence_Parcels	published layer	Approved evidence, not yet operational

2.4 The adapter boundary

`docs/landcare/survey-layer.js` and `docs/landcare/assignment-layer.js` are the only places that talk to ArcGIS. Every source field rename gets absorbed there and nowhere else. That is why `additional_comments`, `additional_notes`, `additional_note`, and `notes` all

normalise to one property, and why `service_date`, `date_of_services`, and `date_services` collapse to one field.

Keep it that way. An ArcGIS URL appearing in `monitoring.js` or `kpi.js` is a defect.

2.5 Where to make a change

To change	Open
How survey evidence is read, normalised, matched	<code>docs/landcare/survey-layer.js</code>
How assignments are read and normalised	<code>docs/landcare/assignment-layer.js</code>
Supervisor map behaviour	<code>docs/landcare/monitoring.js</code>
KPI cards, trends, charts	<code>docs/landcare/kpi.js</code>
Contractor progress and submission	<code>docs/landcare/contractor-overview.js</code>
Survey123 URL, prefill names, evidence layer URL	<code>docs/landcare/survey-submission-config.js</code>
Product styling	<code>docs/landcare/app.css</code>
Nightly VM job and its QA gate	<code>scripts/refresh_landcare_dashboard.ps1</code> , <code>scripts/validate_landcare_daily_refresh.py</code>
Published data build	<code>scripts/build_landcare_web_data.py</code> , <code>scripts/build_landcare_finance_data.py</code>

3 Metrics and the denominator rule

This section is the one to read before changing any number on screen. The full glossary is `docs/landcare-metrics-context.md`.

3.1 Completion

The numerator is the raw count of survey records whose normalised parcel number matches an assignment in the active filter. The denominator is Active assigned. Request Only parcels are excluded from the Active denominator.

Repeated records are kept on purpose, so this is **not** a unique-parcel count. Unique completed parcels exist as a diagnostic only. Map Monitor and the KPI dashboard use the same raw matched numerator, which is why their figures agree.

3.2 Definitions in use

Metric	Definition
Current URA-owned LandCare parcels	Live ArcGIS EPP records tagged LandCare with <code>inventory_type = 'URA Owned'</code>
Active assigned	Active assignment records in the selected period and filter
Request Only	Assignment rows at Request Only maintenance level; assigned inventory, not failure
Complete survey records	Raw survey records matched to an assignment parcel in the filter
Unique completed parcels	Distinct assignment parcel keys with at least one match; diagnostic only
Open active	Active assignment parcel keys with no matched evidence; the follow-up queue

Metric	Definition
Active completion %	Complete survey records over Active assigned; the primary operational KPI
Blended completion %	Complete survey records over assigned total, Request Only included; secondary context
All ArcGIS survey records	Raw survey features for the selected period, including unmatched rows

3.3 Reconciliation rules

Do not compare all ArcGIS survey records directly to complete survey records; the first includes survey-only rows, the second is filtered to assignment matches. Do not compare the current parcel universe to a monthly assignment period; the first is live inventory, the second is a historical export. Treat Request Only as inventory context.

Finance figures come from the LandCare budgeting workbook, not from ArcGIS. The contract parcel count is contract scope and is a different population from either the EPP universe or a monthly assignment slice. Do not mix them in one ratio.

4 Daily operations

4.1 The three-minute morning check

1. Confirm the 4:00 AM upstream pipeline ran and ArcGIS layer freshness moved.
2. Confirm the 7:00 AM task and read `C:\srv\logs\land-care-assurance\daily-refresh-status.json`.
3. Confirm the Pages workflow is green.

If the status JSON is not success, read `failed_stage` and fix only that stage. The full triage tree is in `docs/task-scheduler-vm-operations.md`.

4.2 Logs and gates

Artifact	Path	Purpose
Daily log	<code>C:\srv\logs\land-care-assurance\daily-refresh-YYYY-MM-DD.log</code>	Human troubleshooting
Status JSON	<code>C:\srv\logs\land-care-assurance\daily-refresh-status.json</code>	Success or failure artifact
QA validator	<code>scripts/validate_landcare_daily_refresh.py</code>	Blocks publish on regression or stale manifest
Pages validation	<code>.github/workflows/pages.yml</code>	Data contract check on merge

The Pages workflow does more than deploy. It checks that all routes exist, that the summary JSON, GeoJSON, and refresh manifest agree on counts, and that ownership values stay within URA and Pittsburgh Land Bank. Treat a red Pages workflow as a data contract failure, not a deployment failure.

4.3 Morning executive brief

`.github/workflows/landcare-morning-brief.yml` runs at 9:00 AM Eastern, handling day-light saving by firing at both candidate UTC hours and publishing only in the real window. It sends HTML email through Microsoft 365 Graph when the secrets are present, and otherwise creates an assigned GitHub Issue labelled `landcare-brief`.

After the transfer, set the repository variables `LANDCARE_EMAIL_RECIPIENTS` and `LANDCARE_ISSUE_ASSIGNEE`, then test with the dry-run delivery mode before going live.

5 Replacing Regrid with a URA front end

5.1 Scope it honestly

Regrid provides one thing URA does not yet own: the contractor capture form and its daily export, which lands in `gis.regrid_survey_submissions` and publishes to `gisdb_gis_regrid_surveys`. Assignments, parcel geometry, ownership, the dashboard, and the contractor portal are already URA property.

You are replacing an intake form and an export, not a platform. ArcGIS stays.

5.2 Most of it is already written

Piece	Where	State
Contractor portal, map and list parcel selection	<code>docs/contractor/, contractor-overview.js</code>	Live
Survey123 intake and prefill contract	<code>survey-submission-config.js</code>	Live
Webhook receiver	<code>landcare_survey123_webhook.py</code>	Written, not deployed
Evidence validation and sync	<code>survey123_evidence_sync.py</code>	Written, not deployed
Evidence layer publisher, REST, no ArcGIS Pro licence	<code>publish_landcare_survey_evidence_parcel.py</code>	Written, not deployed
Database migration	<code>sql/20260728_landcare_survey123_evidence_parcel.sql</code>	Not applied

5.3 What is left

Apply the migration. Set the VM environment variables listed in `docs/survey123-landcare-network-setup.md`. Run the receiver behind the approved URA HTTPS host. Register the Survey123 webhook for new records and edits with the `X-LandCare-Webhook-Token` header. Bootstrap the evidence hosted layer once in the LandCare - Published Layers folder and record its item ID. Then run in parallel with Regrid.

5.4 Three gates

Stage	Outcome	Done when
1. Intake	Contractor picks a parcel on the URA map or list, Survey123 opens prefilled with organisation, parcel, address, and period	Map tap and dropdown produce the same prefill for the same OBJECTID; an anonymous browser loads the form without sign-in
2. Review and evidence	URA approves or rejects in a restricted view; only approved records reach the public evidence layer	One approved submission creates exactly one database row and one evidence feature, even after a webhook retry; pending and rejected records appear nowhere public
3. Cutover	Map Monitor and KPI read the URA evidence source instead of Regrid	Two full service periods of parallel running with matching counts, parcel matches, contractors, dates, comments, and photos

5.5 Guardrails, not preferences

Never publish the raw Survey123 point layer or the review QA view on a public map. No publisher token, password, or portal credential goes into GitHub Pages, because Pages content is public. Public views are add-only for submission and query-only for reference. Only approved evidence becomes visible.

5.6 After sign-off

Repoint Map Monitor and KPI, update docs/landcare-metrics-context.md and the tests in the same pull request, then retire the Regrid scheduled task and its credential. Retiring Regrid also removes the browser-automation download step, which is the most fragile part of the current pipeline.

If Survey123 later constrains the contractor experience, replace only the form. Build a custom LandCare form that posts to a protected URA API which validates and writes to PostgreSQL or a secured feature service. Keep the browser adapter contract unchanged: parcel number, period, contractor, service date, image URL, and canonical additional_notes. The application does not need to know which form produced the record.

6 Working with AI agents

This repository was built with agent-assisted development and is set up to keep working that way. The detailed prompt library is `handover/02-agent-playbook.md`. This section is the summary.

AGENTS.md at the repository root is read automatically by Codex and by Claude Code. It carries the source-of-truth rules, the change rules, and the verification requirement. Keep it current. It is the cheapest way to stop an agent from doing the wrong thing.

Every prompt should carry four blocks: the files to read first, the contract (goal, source, metric rule), the commands the agent must actually run, and a task-specific snippet.

6.1 Four failure modes specific to this repository

1. **Denominator drift.** An agent will switch the numerator to unique parcels because it looks more correct. It is not the agreed metric and it changes every published number.
2. **Stale cache-busting.** Modules import each other with `?v=` strings, for example `survey-layer.js?v=20260803-raw-completion-v2`. Change a shared adapter and every importer's version must move together, or browsers serve a mixed bundle.
3. **Hardcoded endpoints.** ArcGIS URLs belong in the two adapter files and in `survey-submission-config.js`, nowhere else.
4. **Hand-editing generated data.** Files under `docs/landcare/data/` come from the VM refresh, so hand edits are silently reverted by the next run.

6.2 Verification before any push

```
python -m unittest discover -s tests
node --test tests/survey-layer.test.mjs
```

`tests/test_handover_contract.py` guards the handover file set and the portability of the design system CSS. Then open the four routes and confirm counts, comments, photos, field notes, filters, and the selected period.

7 Design system

There are three style sources and they are not interchangeable. Knowing which is which saves a great deal of confusion.

Source	What it is	Use it for
<code>docs/landcare/app.css</code>	About 2,800 lines, the live product style, coupled to the ArcGIS JS API 4.30 theme	Changing an existing page
<code>.claude/skills/ura-landcare-design/</code>	The formalised design system extracted from <code>app.css</code> : tokens, 8 components, guidelines, 3 screen templates	Designing anything new, with or without an agent
<code>docs/design-system/executive-bi.css</code>	156 portable lines with its own scoped <code>--bi-*</code> tokens, same palette, different naming	A standalone BI dashboard outside LandCare

The skill's `tokens/colors.css` is an exact extraction of the `:root` block in `app.css`, so the product and the design system cannot drift apart silently. `github.md` inside the skill carries a screen map from each design artifact back to its repo source.

7.1 Invoking it

The design system is installed as a Claude Code skill. Any agent opening this repository discovers it automatically and can be asked to use it by name:

```
/ura-landcare-design
```

It also works as plain reference material. Open `readme.md` inside the skill folder for the full content and visual rules.

7.2 The rules that get checked in review

One brand blue family carries almost everything. Orange means risk or action needed, exclusively. Green means success, target, or complete, exclusively. Gold is a rare tertiary accent. Never more than these four hues plus ink, muted, and line neutrals on a page.

Manrope only, weights 400 to 800, no serif or mono in the UI. Headings are heavy with tight negative letter-spacing; body copy is never bold. A small uppercase kicker precedes every heading.

Flat backgrounds, no photography, no texture, and exactly one gradient reserved for the single featured metric card per band. Hairline 1px borders, two shadow tokens, corner radii of 12, 18, 22, and full pill. Animation limited to short colour transitions on hover.

Never use colour alone; pair every status colour with a text label. Every important number shows its source, period, denominator, and freshness. Normal text contrast at least 4.5:1. Tabs and actions are real buttons or links with `aria-selected`. Tables keep `table`, `th`, and `td`. Layout stacks at 760px and below. Loading, empty, stale, and error states are explicit, because live ArcGIS queries fail in ways static pages do not.

Copy is operational and procedural, never marketing-toned. Sentence case throughout. No emoji, no exclamation points, no filler adjectives. State unavailable data directly rather than showing a vague empty state.

One caution: `tokens/typography.css` imports Manrope from Google Fonts, so the skill's templates need network access. This does not affect `executive-bi.css`, which is deliberately import-free and is asserted so by `tests/test_handover_contract.py`.

8 Ownership, risks, and open decisions

Five areas have distinct owners: source freshness, VM refresh, application code, contractor follow-up, and finance. The escalation table lives in `HANDOVER.md`.

8.1 Open items to decide, not defects

Item	What it means for you
Survey123 evidence path is code-complete but not deployed	Approved photos cannot appear and Regrid stays the only evidence source. The single blocker on the replacement
Survey123 submissions do not change official completion in v1	Whether they should is a governance decision, not a code change
Committed snapshot is dated 2026-07-07 while live ArcGIS moves daily	Static and live numbers differ. Decide whether the fallback stays once the live path is trusted
Morning brief falls back to a GitHub Issue without Microsoft 365 secrets	Set the two repository variables and test with dry-run first
GitHub redirects the old repository path, Pages URLs do not	Every ArcGIS embed, bookmark, and operational link moves to the <code>ura-gis</code> Pages URL by hand

8.2 First week

1. Confirm you can clone, push, and run the tests.
2. Watch two unattended refresh cycles complete with status: success.
3. Run the morning brief manually in dry-run mode and read the artifact.
4. Open all four routes and reconcile one contractor's numbers by hand against the map.
5. Make one harmless documentation change through an agent, end to end, including tests.
6. Decide whether to deploy the Survey123 evidence path this quarter.

9 Appendix

9.1 Repository map

```
land-care-assurance/
  handover/                <- this folder: start here
  AGENTS.md                 <- agent start rules, read automatically
  HANDOVER.md              <- cutover checklist
  .claude/skills/ura-landcare-design/ <- design system skill
  docs/
    monitoring/ kpi/ contractor/ <- the three app routes
    landcare/                <- app JS, CSS, assets, and published data contract
    design-system/           <- portable Executive BI kit
    *.md                     <- architecture, metrics, runbooks
  scripts/                  <- VM daily refresh, data build, brief generation
  sql/                      <- Survey123 evidence migration
  tests/                    <- Python and Node tests
  .github/workflows/        <- Pages deploy and morning brief
```

9.2 Published data contract

Generated daily by the VM refresh into docs/landcare/data/.

File	Contents
all_months.geojson	URA-owned assignment rows with geometry, all periods
latest_month.geojson, latest_month_summary.json	Latest comparable month slice
monthly_metrics.json	Completion rates by month
contractor_monthly.json	Contractor completion by month
kpi_summary.json	Summary metadata and latest metrics
finance_summary.json	Budget and contract totals
refresh_manifest.json	Freshness, counts, survey metadata

9.3 Environment variable names

Names only. Values are VM-local or repository secrets and are never committed.

VM .env: PG_HOST, PG_PORT, PG_DB, PG_USER, PG_PWD.

Survey123 evidence path: LANDCARE_PG_DSN, LANDCARE_SURVEY_WEBHOOK_TOKEN, SURVEY123_FEATURE_LAYER_URL, SURVEY123_ARCGIS_TOKEN, SURVEY123_PUBLIC_ATTACHMENT_LAYER_URL,

LANDCARE_ASSIGNMENT_HISTORY_LAYER_URL, LANDCARE_SURVEY_EVIDENCE_ENABLED, LANDCARE_SURVEY_EVIDENCE_AGOL_ITEM_ID, LANDCARE_SURVEY_EVIDENCE_LAYER_URL, LANDCARE_SURVEY_EVIDENCE_ARCGIS_TOKEN.

GitHub Actions secrets: M365_TENANT_ID, M365_CLIENT_ID, M365_CLIENT_SECRET, M365_SENDER_UPN. Repository variables: LANDCARE_EMAIL_RECIPIENTS, LANDCARE_ISSUE_ASSIGNEE.

9.4 Command reference

```
python -m unittest discover -s tests           # Python tests
node --test tests/survey-layer.test.mjs       # survey adapter tests
python handover/build/build_report.py         # rebuild this PDF
```

On the VM:

```
.\scripts\refresh_landcare_dashboard.ps1 ^
-RepoRoot C:\srv\GISWebApp\land-care-assurance
.\.venv\Scripts\python.exe scripts\validate_landcare_daily_refresh.py
```

9.5 Further reading

- docs/landcare-architecture.md for source and runtime architecture
- docs/landcare-metrics-context.md for the full metric glossary
- docs/landcare-submission-and-evidence-flow.md for the contractor intake contract
- docs/survey123-landcare-network-setup.md for Survey123 and webhook configuration
- docs/task-scheduler-vm-operations.md for the VM runbook and failure triage
- docs/github-handover-runbook.md for the repository and Pages cutover